Substitution and Rearranging Formulae

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

KEY WORDS substitute *(whakakapi)* · expression *(kīanga taurangi)* · formula *(tātai)* · subject

A Substituting positive numbers ●○○

Remember: Two letters side by side are multiplied: ab means $a \times b$, and $3a$ means $3 \times a$.

1. Find the value of each expression when $a = 3$, $b = 5$ and $c = 2$.

a) $2a + b$ = _____ **b)** ab = _____ **c)** $b^2 - a$ = _____ **d)** $3(a + b)$ = _____

e) abc = _____ **f)** $\dfrac{b + 7}{a}$ = _____ **g)** $2a^2$ = _____ **h)** $a^2 + c^2$ = _____

i) $4b - 3c$ = _____ **j)** $\dfrac{ab + 1}{c}$ = _____ **k)** $(a + c)^2$ = _____ **l)** $c^3 - b$ = _____

B Negative numbers and powers ●●○

Watch out: Put brackets around a negative number when you substitute it: $(-3)^2 = 9$, not -9.

2. Find the value of each expression when $p = -2$, $q = 4$ and $r = -3$.

a) $p + q$ **b)** pq **c)** $q - r$ **d)** p^2

e) pr **f)** $2p + 3r$ **g)** $r^2 - q$ **h)** $2p^2 + r$

i) pqr **j)** $\dfrac{q - p}{r}$ **k)** $q^2 - p^2$ **l)** $3r^2 - 2q$

C Using formulae ●●○

3. Substitute into each formula.

a) $v = u + at$. Find v when $u = 5$, $a = 3$ and $t = 4$.

b) $A = \dfrac{1}{2}bh$. Find A when $b = 8$ and $h = 5$.

c) Celsius to Fahrenheit: $F = 1.8C + 32$. Find F when and $C = 25$.

d) $s = ut + \dfrac{1}{2}at^2$. Find s when $u = 4$, $a = 6$ and $t = 3$.

e) $P = 2(l + w)$. Find P when $l = 7.5$ and $w = 4$.

f) Area of a trapezium: $A = \dfrac{h(a + b)}{2}$. Find A when $h = 6$, $a = 5$ and $b = 9$.

g) $E = \dfrac{1}{2}mv^2$. Find E when $m = 4$ and $v = 5$.

h) A plumber charges C dollars for h hours: $C = 40 + 25h$. Find C when and $h = 3$.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Rearranging: one step ●●○

Remember: To make a letter the subject, undo what has been done to it. Do the same to both sides.

4. Make the letter in square brackets the subject of the formula.

a) $y = x + 5 \quad [x]$

b) $A = lw \quad [w]$

c) $P = 4s \quad [s]$

d) $v = \dfrac{d}{t} \quad [d]$

e) $y = x - c \quad [x]$

f) $C = 2\pi r \quad [r]$

g) $F = ma \quad [a]$

h) $m = \dfrac{k}{5} \quad [k]$

i) $V = lwh \quad [h]$

E Rearranging: two steps ●●●

Remember: Undo adding and subtracting first. Then undo multiplying and dividing.

5. Make the letter in square brackets the subject of the formula.

a) $y = mx + c \quad [x]$

b) $v = u + at \quad [t]$

c) $A = \dfrac{1}{2}bh \quad [h]$

d) $P = 2l + 2w \quad [l]$

e) $F = 1.8C + 32 \quad [C]$

f) $y = \dfrac{x}{3} - 2 \quad [x]$

g) $y = 5(x + a) \quad [x]$

h) $T = \dfrac{a + b}{2} \quad [a]$

i) $y = k - 3x \quad [x]$

★ Extension

6. These formulae have squares and square roots. Make the letter in square brackets the subject.

a) $A = \pi r^2 \quad [r]$

b) $E = \dfrac{1}{2}mv^2 \quad [v]$

c) $v^2 = u^2 + 2as \quad [s]$

d) $y = \sqrt{x + a} \quad [x]$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Substitution and Rearranging Formulae

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS substitute *(whakakapi)* · expression *(kīanga taurangi)* · formula *(tātai)* · subject

A Substituting positive numbers ●○○

Remember: Two letters side by side are multiplied: ab means $a \times b$, and $3a$ means $3 \times a$.

1. Find the value of each expression when $a = 4$, $b = 6$ and $c = 3$.

a) $3a + b$ = _____ **b)** ac = _____ **c)** $b^2 - c$ = _____ **d)** $2(a + c)$ = _____

e) abc = _____ **f)** $\dfrac{b + 6}{c}$ = _____ **g)** $3c^2$ = _____ **h)** $a^2 + b^2$ = _____

i) $5a - 2b$ = _____ **j)** $\dfrac{ab + 3}{c}$ = _____ **k)** $(b - c)^2$ = _____ **l)** $c^3 - a$ = _____

B Negative numbers and powers ●●○

Watch out: Put brackets around a negative number when you substitute it: $(-3)^2 = 9$, not -9.

2. Find the value of each expression when $p = -3$, $q = 5$ and $r = -2$.

a) $p + q$ **b)** pr **c)** $q - p$ **d)** r^2

e) pq **f)** $3p + 2r$ **g)** $p^2 - q$ **h)** $2r^2 + p$

i) pqr **j)** $\dfrac{q - p}{r}$ **k)** $q^2 - r^2$ **l)** $2p^2 - 3q$

C Using formulae ●●○

3. Substitute into each formula.

a) $v = u + at$. Find v when $u = 8$, $a = 2$ and $t = 6$.

b) $A = \dfrac{1}{2}bh$. Find A when $b = 10$ and $h = 7$.

c) Celsius to Fahrenheit: $F = 1.8C + 32$. Find F when and $C = 35$.

d) $s = ut + \dfrac{1}{2}at^2$. Find s when $u = 5$, $a = 4$ and $t = 3$.

e) $P = 2(l + w)$. Find P when $l = 6.5$ and $w = 3$.

f) Area of a trapezium: $A = \dfrac{h(a + b)}{2}$. Find A when $h = 4$, $a = 7$ and $b = 11$.

g) $E = \dfrac{1}{2}mv^2$. Find E when $m = 6$ and $v = 3$.

h) An electrician charges C dollars for h hours: $C = 60 + 35h$. Find C when and $h = 4$.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Rearranging: one step ●●○

Remember: To make a letter the subject, undo what has been done to it. Do the same to both sides.

4. Make the letter in square brackets the subject of the formula.

a) $y = x + 8 \quad [x]$

b) $A = bh \quad [h]$

c) $P = 6s \quad [s]$

d) $s = \dfrac{d}{t} \quad [d]$

e) $y = x - k \quad [x]$

f) $C = \pi d \quad [d]$

g) $V = IR \quad [I]$

h) $n = \dfrac{t}{4} \quad [t]$

i) $V = lwh \quad [l]$

E Rearranging: two steps ●●●

Remember: Undo adding and subtracting first. Then undo multiplying and dividing.

5. Make the letter in square brackets the subject of the formula.

a) $y = mx - c \quad [x]$

b) $v = u + at \quad [a]$

c) $A = \dfrac{1}{2}bh \quad [b]$

d) $P = 2l + 2w \quad [w]$

e) $C = 40 + 25h \quad [h]$

f) $y = \dfrac{x}{4} + 3 \quad [x]$

g) $y = 2(x - b) \quad [x]$

h) $M = \dfrac{p + q}{2} \quad [q]$

i) $y = n - 5x \quad [x]$

★ Extension

6. These formulae have squares and square roots. Make the letter in square brackets the subject.

a) $V = \pi r^2 h \quad [r]$

b) $A = 4\pi r^2 \quad [r]$

c) $v^2 = u^2 + 2as \quad [a]$

d) $y = \sqrt{x - b} \quad [x]$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.